Generating Videos with Dynamics-aware Implicit Generative Adversarial Networks

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Project page (videos): https://sihyun-yu.github.io/digan
Paper: https://arxiv.org/abs/2202.10571
Code: https://github.com/sihyun-yu/digan

TL;DR. We make video generation scalable leveraging implicit neural representations

Motivation

Videos are too complex… Prior generative models were not scalable ☹

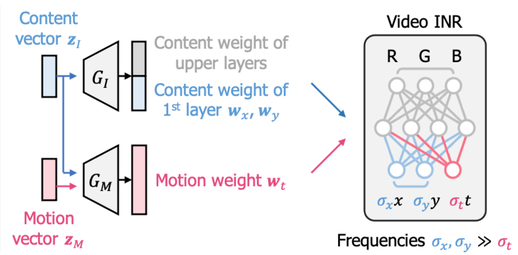
Idea. Interpret videos as implicit neural representations (INRs) - also called as continuous neural fields - instead of discrete spatiotemporal grids

$(x, y, t) \rightarrow$  $\rightarrow (R, G, B)$

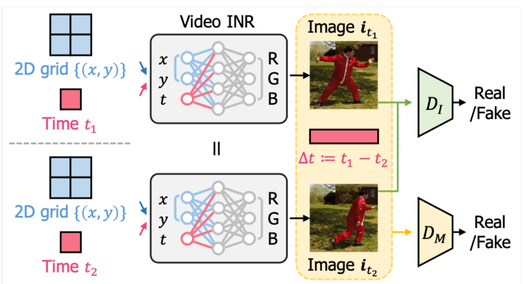
DIGAN: Dynamics-aware Implicit GAN

DIGAN follows the generative adversarial network (GAN) framework

Generator. Generate INR weights (instead of videos directly)
- Synthesize motion weight w_t in addition to the content weights
- Use smaller frequency for time than space (temporal coherency)

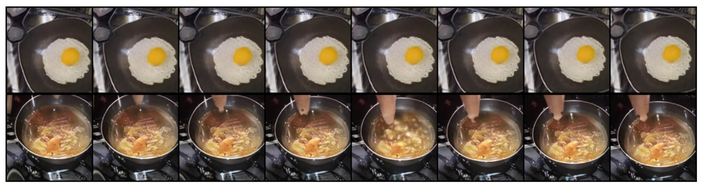


Discriminator. Use two discriminators: image D_I and motion D_M
- Motion D_M uses a pair of image and their time difference $(i_{t_1}, i_{t_2}, t_1 - t_2)$ as input, instead of the full sequence of images (much efficient)



Qualitative & Quantitative Results

DIGAN significantly outperforms the prior arts (e.g., MoCoGAN-HD)



(a) UCF-101

Method	IS (↑)	FVD (↓)
Train split		
VGAN	8.31±.09	-
TGAN	11.85±.07	-
MoCoGAN	12.42±.07	-
ProgressiveVGAN	14.56±.05	-
LDVD-GAN	22.91±.19	-
VideoGPT	24.69±.30	-
TGANv2	28.87±.67	1209±28
DIGAN (ours)	**29.71**±.53	**655**±22
Train+test split		
DVD-GAN	27.38±.53	-
MoCoGAN-HD	32.36	838
DIGAN (ours)	**32.70**±.35	**577**±21

(b) Sky

Method	FVD (↓)	KVD (↓)
MoCoGAN-HD	183.6±5.2	13.9±0.7
DIGAN (ours)	**114.6**±4.3	**6.8**±0.5

(c) TaiChi

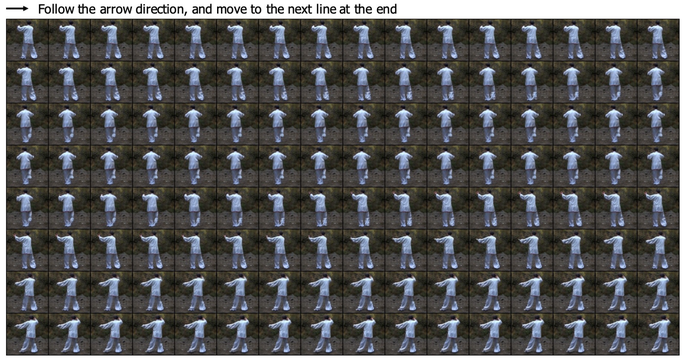
Method	FVD (↓)	KVD (↓)
MoCoGAN-HD	144.7±6.0	25.4±1.9
DIGAN (ours)	**128.1**±4.9	**20.6**±1.1

(d) Kinetics-food

Method	FVD (↓)	KVD (↓)
MoCoGAN-HD	430.4±29.9	276.0±50.7
DIGAN (ours)	**313.3**±36.9	**183.0**±40.3

Long Video Generation

DIGAN generates 5.3x longer (256 frames) videos than the prior arts



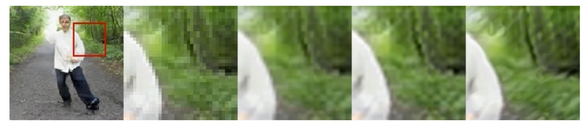
Spatio-temporal Inter- & Extra-polation

DIGAN can interpolate & extrapolate frames both spatially & temporally

Temporal extrapolation. Yellow frames are temporally extrapolated



Space interpolation. Video super-resolution for free

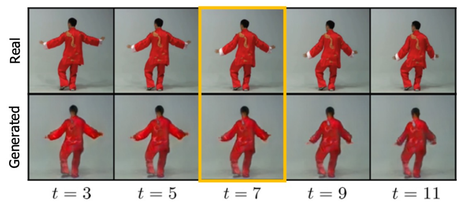


See the paper for more (e.g., time interpolation) results

Non-autoregressive Generation

DIGAN can generate frames of arbitrary time (no sequential order): can predict past (not only future) frames from conditioned frames
- Find the corresponding latent $\{z_I, z_M\}$ from conditioned frames

Backward prediction. Yellow frame is the conditioned frame



See the paper for more (e.g., intermediate prediction) results

Training & Inference Efficiency

DIGAN is much efficient than the prior arts in many aspects, e.g.,
- 4.6x less FLOPS than MoCoGAN-HD with 2x less training time